# Pradeep Somasundaram

McKinney, Texas, United States | pradeep.somasundaram1501@gmail.com | (571) 259-4793 | github.com/PradeepSomasundaram1512 | linkedin.com/in/pradeep-somasundaram-835230192 | pradeepsomasundaram.netlify.app

## PROFESSIONAL SUMMARY

AI and data engineer who ships production data pipelines and LLM systems and audits them with evaluation harnesses. Migrated legacy SQL Server/PL-SQL workloads to GCP at 10M+ records monthly and about 500 GB to Databricks, and built Azure data workflows with Data Factory and Synapse. Ran a 7-node LangGraph agent for nine months and found 71.3% of its decisions were silent API fallbacks. One paper accepted at ICAIET-2027 and one submitted to IEEE AIEI 2027. Google Cloud Data Engineer and ML Engineer, AWS Generative AI Professional, and Microsoft certified.

## SKILLS

**AI and LLM Systems:** LLMs, Agentic AI, LangGraph, LangChain, Model Context Protocol (MCP), Tool/Function Calling, RAG, Embeddings, Pinecone, Evaluation Harnesses, LangSmith, MLflow, Claude Code, Cursor
**Data Engineering:** Python, SQL, Scala, PySpark, Spark SQL, Databricks, Delta Lake, Kafka, Airflow, GCP BigQuery and Dataflow, Azure Data Factory, ADLS, Synapse, ETL/ELT, Medallion Architecture, Incremental Processing, Data Quality
**Machine Learning:** XGBoost, LightGBM, TensorFlow, LSTM, Time-Series Modeling, Anomaly Detection, Feature Engineering, Model Evaluation, Statistical Testing (Matched Placebos, Block Bootstrap)
**Backend and Cloud:** FastAPI, REST, gRPC, TypeScript, React, Docker, Kubernetes, AKS, CI/CD, GitHub Actions, Azure DevOps, AWS S3, Glue, Lambda, IAM, Monitoring, Observability

## WORK EXPERIENCE

**Cognizant Technology Solutions | Agentic AI & MLOps Lead** **Feb 2026 - Present**
- Led technical design of an agentic-AI automation initiative, delivering a proof-of-concept 5-agent LangGraph workflow (intake to RCA).
- Modernized SQL Server and PL/SQL workloads on GCP BigQuery and Dataflow at 10M+ records monthly, cutting latency about 30%.
- Contributed to MLOps foundations: Docker, Kubernetes and AKS deployments, Terraform-style IaC, and monitoring that catches failed loads.
- Built Azure workflows (Data Factory, ADLS, Synapse), Airflow orchestration, and Bronze/Silver/Gold layers with data-quality checks.

**Beauty Manufacturing Solutions Corp (BMSC) | Agentic AI Engineer / Data Engineer** **Sep 2025 - Feb 2026**
- Built PySpark and Spark SQL pipelines on Databricks, migrating ~500 GB of SQL Server data to AWS S3 and Delta Lake, 2M+ records monthly.
- Handled late-arriving records with incremental loads and Delta Lake MERGE upserts, with deduplication and validation before promotion.
- Owned migration workstreams from design through deployment and production stabilization across AWS, Databricks, and consumers.
- Shipped RAG-based LLM workflows grounded in 4 enterprise systems using embeddings and retrieval orchestration.

**Shree Kay Vee Automation | Data Scientist** **Mar 2020 - Jul 2023**
- Cut data-processing latency by 40% by building Python and SQL ingestion pipelines for telemetry from 100+ industrial assets.
- Developed LSTM anomaly-detection and predictive-maintenance models across 100+ assets with validated features.
- Led a 3-member technical team delivering AI and data engineering solutions across development, integration, and rollout.
- Integrated 2 data domains, telemetry and enterprise, through APIs and cloud services running on Linux.

## PROJECTS

**AI Trading Platform | React, TypeScript, Python, Node.js, LangGraph, RAG, AWS, GCP** **Jan 2026 - Sep 2026**
- Built a full-stack platform with a React/TypeScript front end, Python/Node.js APIs, live market and news integrations, and alerts.
- Deployed a 7-node LangGraph agent logging 3,507 live decisions, and found 71.3% were silent API-quota fallbacks.
- Built an evaluation harness where matched placebos erased an apparent +1.71% ten-day return and exposed 82% stated confidence vs 45% wins.

**Cricket Win Probability and Player Impact | LightGBM, Elo, FastAPI, React**
- Built per-innings LightGBM win-probability models with leakage-safe pre-match Elo on 3,364 T20I matches, tested chronologically.
- Reached test log loss of 0.554 (first innings) and 0.349 (chase) against a 0.693 baseline, served through FastAPI and React.

**Real-Time Anomaly Detection and Fraud Detection | Kafka, Spark Streaming, XGBoost**
- Built a Kafka and Spark Streaming pipeline for stock data with ML anomaly detection using XGBoost and deep learning.
- Designed the pipeline for millions of events per second, with streaming ingestion and anomaly scoring.
- Built fraud detection with Isolation Forest and Autoencoders on a Kafka and AWS workflow, with about 30% fewer fraudulent occurrences.

**Problematic Internet Use Prediction | XGBoost, LightGBM, BERT, LSTM**
- Compared XGBoost, LightGBM, CatBoost, BERT, LSTM, and Transformers on behavioral and actigraphy data; LightGBM led at QWK 0.52.

**AgriDroneAI | Autonomous Drone Operations Agent | Python, GitHub Actions, PX4**
- Built a self-operating Python agent with safety gate, kill switch, and audit trail, saving about 80% water in simulation.
- Wrote 8 evaluation and benchmark scripts covering stress model, sensor-fault recall, wind, and robustness before deployment.
- Shipped 220+ tests behind an 85% coverage gate, with daily GitHub Actions cycles that open incidents on failure.

## PUBLICATIONS

**Does the LLM Add Alpha? A Contamination-Free Forward Audit of a Deployed Multi-Agent LLM Trading System** **Submitted, AIEI 2027**
- Proposed the Forward Audit Protocol (matched placebo, block bootstrap, calibration audit), with code and data released publicly.
- Showed the LLM's selection alpha was statistically indistinguishable from zero at every horizon, with effect sizes stated.

**Reliable Evaluation of ARIMA-LSTM Hybrids for Remaining Useful Life Prediction** **Accepted, ICAIET-2027**
- Tested on 4 NASA C-MAPSS datasets over 20+ seeds, where a moving average beat causal ARIMA on 3 of 4.

**HERA-PdM: Uncertainty-Gated Edge-Cloud Predictive Maintenance** **Submitted, ICAIET 2027**
- Conformalized quantile regression cut critical moments left without action to 0.6% vs 49.5% for thresholds on FEMTO bearings.

## EDUCATION

**George Washington University | Master of Science, Data Science** **Aug 2023 - May 2025**

## CERTIFICATIONS & ACHIEVEMENTS

AWS Certified Generative AI Developer - Professional | Google Cloud Professional Data Engineer | Google Cloud Professional ML Engineer | Microsoft Agentic AI Business Solutions Architect Expert | AWS Certified Cloud Architect